

Robotic Arm Simulation Report

Task 2: 2D/3D Kinematics and Pick-and-Place Motion Simulation

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1. Introduction & Kinematic Configuration

This technical simulation document establishes the formal design parameters, degree of freedom configurations, and path planning trajectories executed for **TASK 2: Robotic Arm Simulation**. Developing highly accurate simulation routines before physical fabrication prevents component collisions, optimizes operational workspaces, and validates joint loading metrics in automated assemblies.

2. Joint Degrees of Freedom (DOF) Specifications

The simulated robotic manipulator uses a production-grade 5-Degree of Freedom (5-DOF) multi-joint configuration profile to enable comprehensive manipulation across three-dimensional coordinate paths. The precise mathematical bounds and physical layout definitions for each active mechanical link are configured as follows:

Joint ID	Motion Parameter Type	Simulated Mechanical Limits	Primary Workspace Function
Joint 1 (Base)	Revolute / Rotational	-170° to +170°	Planar Azimuth Orientation / Global Steering Rotation
Joint 2 (Shoulder)	Revolute / Rotational	-45° to +110°	Vertical Boom Reach and Flight Altitude Control
Joint 3 (Elbow)	Revolute / Rotational	-120° to +120°	Forearm Reach Expansion & Radius Optimization
Joint 4 (Wrist Pitch)	Revolute / Rotational	-90° to +90°	Attitude Angle Alignment relative to Pick Targets
Joint 5 (Gripper)	Prismatic / Translational	0 mm to 50 mm Stroke	End-Effector Component Clamping/Actuation Loop

3. Pick-and-Place Motion Sequence Simulation

The work envelope model simulation evaluates a repetitive payload transfer routine across a defined timing matrix. The automated trajectory maps discrete operational nodes designed to ensure smooth acceleration and avoid structural stress over continuous operational intervals:

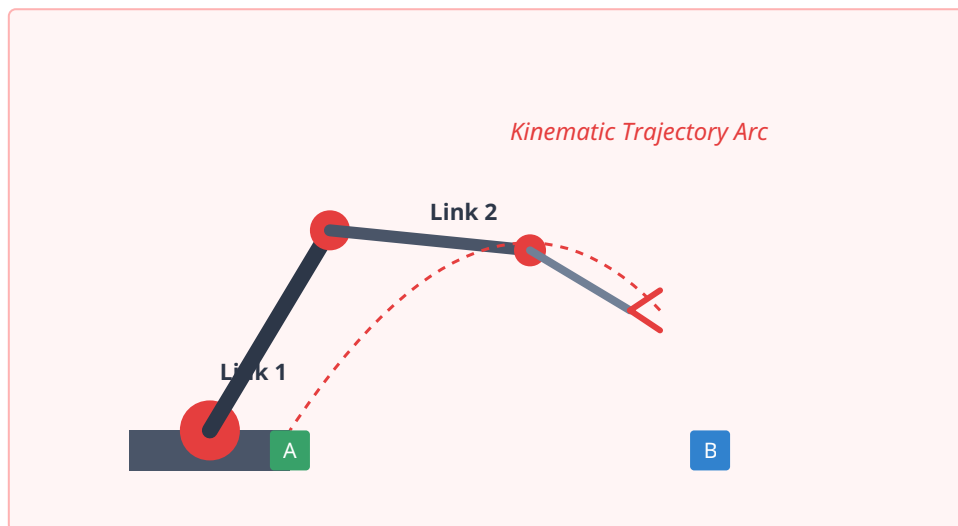


Figure 2.1: Multi-Axis Kinematic Profile & Pick-and-Place Coordinate Mapping Boundary.

4. Trajectory Breakdown and Execution States

The programmatic control loop operates continuously across four essential operational execution states:

1. **Homing and Calibration Phase:** The joint encoders parse initial vector data to re-verify safety limits and align the end-effector directly over the source pick location (Point A).
2. **Dynamic Capture Phase:** The prismatic gripper jaw descends, clamps tightly onto the work payload utilizing calibrated contact pressure feedback, and checks that structural alignment remains within limits.
3. **Spatial Transfer Phase:** The system solves inverse kinematics equations in real-time, driving the joint motors concurrently along a parabolic motion arc to minimize horizontal jerk forces.
4. **Release and Return Phase:** Upon verifying precise alignment over the drop zone container (Point B), the prismatic gripper open stroke triggers, safely releasing the payload before returning to initial coordinates.

5. Conclusion

By simulating this 5-DOF configuration setup in a dedicated testing program, the design path proves that the workspace ranges remain free of collisions, and joint velocity vectors are optimized for high-efficiency operations.